Lecture Notes

# Lecture 4

## Matrices of Linear Maps; Change of Basis

In Lecture 3 we proved an important fact: a linear map is completely determined by the images of the basis vectors. If we also fix a basis of the codomain, we can write each of those images as a coordinate vector. Placing these coordinate vectors side by side as columns gives a *matrix*. In this lecture we turn the informal correspondence "linear map $\leftrightarrow$ matrix" into a precise isomorphism. Three things follow. First, we do not *choose* the rule for multiplying matrices: it is exactly the rule that composition of linear maps demands. Second, an abstract space of linear maps is simply a space of matrices. Third, when we change basis, the matrix of the same operator changes, and the new and old matrices are related by a *similarity* transformation $A' = P^{-1}AP$.

For physics, this dependence on the basis is what matters most. In finite-dimensional (discrete) models, choosing a basis means choosing a coordinate *representation*. Discretized position, momentum and energy models are examples. Continuous position or momentum representations require the Hilbert-space machinery that we study in Lectures 13–14. In later lectures we will choose bases in which an operator's matrix is as simple as possible. The simplest is a diagonal matrix — that will be the goal of the *spectral theorem*.

**Learning objectives**

By the end of this lecture you will be able to:

- Build the matrix $\mathcal{M}(T)$ in chosen bases and use $[Tv] = \mathcal{M}(T)[v]$.
- Explain why composition of maps is matrix multiplication.
- Decide whether a map is invertible, and classify spaces up to isomorphism by dimension.
- Change basis and recognize trace and determinant as similarity invariants.

### 1 The matrix of a linear map

Fix a basis $\mathcal{B} = (v_1, \dots, v_n)$ of $V$ and a basis $C = (w_1, \dots, w_m)$ of $W$.

**Definition 4.1 (Matrix of a linear map).** The *matrix* of a linear map $T \in \mathcal{L}(V, W)$ with respect to the bases $\mathcal{B}$ and $C$ is the $m \times n$ matrix $\mathcal{M}(T) = (A_{jk})$ whose $k$-th column holds the coordinates of the vector $Tv_k$ in the basis $C$:

$$Tv_k = A_{1k}w_1 + \cdots + A_{mk}w_m = \sum_{j=1}^{m} A_{jk}\, w_j.$$

When the bases are not clear from context, we will write $\mathcal{M}\big(T, \mathcal{B}, C\big)$.

In other words, *the $k$-th column is the image of the $k$-th basis vector under $T$, written in coordinates.* When the map is $\mathbb{F}^n \to \mathbb{F}^m$ and the bases are the standard ones, the $k$-th column is simply $Te_k$.

**Example 4.2 (A map $\mathbb{F}^2 \to \mathbb{F}^3$).** Let $T(x, y) = (x + 3y,\ 2x + 5y,\ 7x + 9y)$. The images of the standard basis vectors are $T(1, 0) = (1, 2, 7)$ and $T(0, 1) = (3, 5, 9)$. Writing them as columns

gives

$$\mathcal{M}(T) = \begin{pmatrix} 1 & 3 \\ 2 & 5 \\ 7 & 9 \end{pmatrix}.$$

◁

**Example 4.3 (Rotation of the plane).** The rotation $R_\theta$ of the plane $\mathbb{R}^2$ by an angle $\theta$ moves the basis vectors as follows: $e_1 \mapsto (\cos\theta, \sin\theta)$ and $e_2 \mapsto (-\sin\theta, \cos\theta)$. Hence

$$\mathcal{M}(R_\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

The composition of two rotations corresponds to the product of their matrices. The addition formulae for the sine and cosine of a sum of two angles follow from the equality

$$\mathcal{M}(R_\alpha)\mathcal{M}(R_\beta) = \mathcal{M}(R_{\alpha+\beta}).$$

◁

**Example 4.4 (Differentiation on $\mathcal{P}_3$).** Let $D \in \mathcal{L}(\mathcal{P}_3(\mathbb{R}))$ be the differentiation operator $Dp = p'$, and take the standard basis $1, t, t^2, t^3$. The derivatives of the basis vectors are $D1 = 0$, $Dt = 1$, $Dt^2 = 2t$, $Dt^3 = 3t^2$. Writing them in coordinates and placing them as columns gives

$$\mathcal{M}(D) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Each column holds the coordinates of the derivative of one basis vector. Above the main diagonal stand the exponents $1, 2, 3$. ◁

**Example 4.5 (The matrix depends on the basis).** On $\mathcal{P}_2(\mathbb{R})$ the matrix of differentiation in the standard basis is $\mathcal{M}(D) = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{smallmatrix}\right)$. Now take another basis $\mathcal{B}' = (1,\ 1+t,\ 1+t+t^2)$ and write the derivatives of its vectors in coordinates with respect to this basis. Since $D1 = 0$, the first column is zero. Next, $D(1+t) = 1$, and the coordinates of this polynomial in the basis $\mathcal{B}'$ are $(1, 0, 0)$.

That leaves $D(1+t+t^2) = 1 + 2t$. We seek scalars $a, b, c$ such that

$$1 + 2t = a \cdot 1 + b(1+t) + c(1+t+t^2) \quad \Longrightarrow \quad (1, 2, 0) = a(1,0,0) + b(1,1,0) + c(1,1,1).$$

We write the basis vectors into the columns of a matrix, the polynomial $1 + 2t$ on the right-hand side, and reduce the augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} \textcircled{1} & 0 & 0 & -1 \\ 0 & \textcircled{1} & 0 & 2 \\ 0 & 0 & \textcircled{1} & 0 \end{array}\right] \Longrightarrow\ a = -1, \quad b = 2, \quad c = 0.$$

So the coordinates of this polynomial in the basis $\mathcal{B}'$ are $(-1, 2, 0)$. Placing all three coordinate vectors as columns gives

$$\mathcal{M}(D, \mathcal{B}') = \begin{pmatrix} 0 & 1 & -1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Same operator, different matrix. The section on change of basis explains exactly how the two matrices are related. ◁

The definition of the matrix speaks only about the images of the basis vectors. From them, by linearity, we obtain the image of any vector. Recall that we write $[v]_{\mathcal{B}}$ for the coordinate vector of $v$ in the basis $\mathcal{B}$. Coordinate vectors satisfy the identity that makes matrices so useful for computation.

**Theorem 4.6 (A matrix acts on coordinates).** For every vector $v \in V$,

$$[Tv]_C = \mathcal{M}(T)\,[v]_{\mathcal{B}}.$$

*Proof.* Expand the vector $v$ in the basis: $v = \sum_k b_k v_k$, so $[v]_{\mathcal{B}} = (b_1, \dots, b_n)$. Since $T$ is linear,

$$Tv = \sum_k b_k\, Tv_k = \sum_k b_k \sum_j A_{jk} w_j = \sum_j \Big(\sum_k A_{jk} b_k\Big) w_j.$$

Hence the $j$-th coordinate of $Tv$ in the basis $C$ is $\sum_k A_{jk} b_k$. This is exactly the $j$-th entry of the product $\mathcal{M}(T)\,[v]_{\mathcal{B}}$. ∎

The commutative diagram in Figure 1 illustrates this theorem. Whether we first apply $T$ and then pass to coordinates, or first pass to coordinates and then multiply by $\mathcal{M}(T)$, the result is the same.

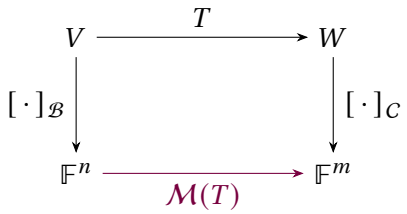



Figure 1: The matrix is the map written in coordinates: passing to coordinates and multiplying by $\mathcal{M}(T)$ gives the same result as applying $T$ and only then passing to coordinates.

**Example 4.7 (Computing $Tv$ through the matrix).** Take the differentiation operator $D$ on $\mathcal{P}_3(\mathbb{R})$ (Example 4.4) and the polynomial $p(t) = 2 + t - t^2 + 4t^3$. Its coordinate vector is $[p] = (2, 1, -1, 4)$. Then

$$[Dp] = \mathcal{M}(D)\,[p] = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 12 \\ 0 \end{pmatrix},$$

that is, $Dp = 1 - 2t + 12t^2$. Differentiating directly gives the same derivative. The matrix has turned differentiation into arithmetic. ◁

## 2 Composition becomes matrix multiplication

Matrices are added and multiplied by scalars entrywise. With these operations the map $\mathcal{M}$ respects the linear structure.

**Proposition 4.8 (The map $\mathcal{M}$ is linear).** For all maps $S, T \in \mathcal{L}(V, W)$ and scalars $a \in \mathbb{F}$,

$$\mathcal{M}(S + T) = \mathcal{M}(S) + \mathcal{M}(T) \quad \text{and} \quad \mathcal{M}(aT) = a\,\mathcal{M}(T).$$

*Proof.* Consider the $k$-th column of the matrix $\mathcal{M}(S+T)$. It holds the coordinates of the vector $(S+T)v_k = Sv_k + Tv_k$ in the basis $C$. The coordinate map is linear, so this column equals the sum of the $k$-th columns of $\mathcal{M}(S)$ and $\mathcal{M}(T)$. The case of multiplication by a scalar is proved in the same way. ∎

What corresponds to the *composition* of maps in the language of matrices? The answer determines how matrices must be multiplied.

**Theorem 4.9 (Composition → multiplication).** Let $T \in \mathcal{L}(U,V)$ and $S \in \mathcal{L}(V,W)$, with bases fixed in all three spaces. Then $\mathcal{M}(ST) = \mathcal{M}(S)\,\mathcal{M}(T)$. Here the product of an $m \times p$ matrix $B = (B_{ij})$ and a $p \times n$ matrix $A = (A_{jk})$ is the $m \times n$ matrix with entries

$$(BA)_{ik} = \sum_{j=1}^{p} B_{ij} A_{jk}. \tag{4.1}$$

*Proof.* Let $\mathcal{B} = (u_1, \dots, u_n)$ be a basis of $U$, $\mathcal{M}(T) = (A_{jk})$ and $\mathcal{M}(S) = (B_{ij})$. Apply first $T$ and then $S$ to the $k$-th basis vector:

$$(ST)u_k = S\Big(\sum_j A_{jk} v_j\Big) = \sum_j A_{jk}\, Sv_j = \sum_j A_{jk} \sum_i B_{ij} w_i = \sum_i \Big(\sum_j B_{ij} A_{jk}\Big) w_i.$$

So the $i$-th coordinate of $(ST)u_k$ is $\sum_j B_{ij}A_{jk}$. This coordinate is the entry of $\mathcal{M}(ST)$ in position $(i,k)$, so we obtain exactly the formula (4.1). ∎

So the "row times column" rule for multiplying matrices is not an arbitrary convention. It is the only rule under which the product of matrices corresponds to the composition of maps. Hence matrices inherit every property of matrix multiplication — associativity, distributivity, failure to commute — from the corresponding property of composing maps.

**Example 4.10 ($\mathcal{M}(D^2) = \mathcal{M}(D)^2$).** Let $D$ be the differentiation operator on $\mathcal{P}_3(\mathbb{R})$, as in Example 4.4. The operator $D^2$ sends the basis vectors $1, t, t^2, t^3$ to $0, 0, 2, 6t$. Multiplying the matrix $\mathcal{M}(D)$ by itself gives the same result:

$$\mathcal{M}(D)^2 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \mathcal{M}(D^2).$$

The band of nonzero entries has moved up by one position, and its entries 2 and 6 are products of consecutive exponents. ◁

**Example 4.11 (Multiplication corresponds to composition).** Let

$$T(x,y) = (x+y,\ y) \quad \text{and} \quad S(x,y) = (2x,\ x-y)$$

be operators on $\mathbb{R}^2$. Their matrices in the standard basis are

$$\mathcal{M}(T) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \mathcal{M}(S) = \begin{pmatrix} 2 & 0 \\ 1 & -1 \end{pmatrix}.$$

The composition is $(ST)(x,y) = S(x+y,y) = (2x+2y,\ x)$, so

$$\mathcal{M}(ST) = \begin{pmatrix} 2 & 2 \\ 1 & 0 \end{pmatrix}.$$

Multiplying the matrices gives

$$\mathcal{M}(S)\mathcal{M}(T) = \begin{pmatrix} 2 & 0 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 1 & 0 \end{pmatrix}.$$

The product agrees with the matrix of the composition, as Theorem 4.9 states. ◁

The map $\mathcal{M}$ from $\mathcal{L}(V,W)$ to the matrix space $\mathbb{F}^{m\times n}$ is linear. It is injective: if the matrix of a map is zero, then that map sends every basis vector to the zero vector, so the map itself is zero. The map $\mathcal{M}$ is also surjective: any matrix prescribes images of the basis vectors, and in Lecture 3 we saw that such images define a unique linear map. Hence $\mathcal{M}$ is a bijective linear map.

A bijective linear map is called an *isomorphism*, and spaces between which an isomorphism exists are called *isomorphic* (written $V \cong W$). We met the term briefly in Lecture 3's subsection on quotient spaces. The next section defines an isomorphism through invertibility and proves that the two definitions agree. So $\mathcal{M}$ is an isomorphism. This lets us compute the dimension promised in Lecture 3.

**Corollary 4.12 ($\mathcal{L}(V,W) \cong \mathbb{F}^{m\times n}$).** The map $\mathcal{M}$ is an isomorphism from $\mathcal{L}(V,W)$ onto $\mathbb{F}^{m\times n}$. Hence

$$\dim \mathcal{L}(V,W) = (\dim V)(\dim W).$$

**Proposition 4.13 (Matrix rank is map rank).** For any choice of bases, the dimension $\dim \operatorname{im} T$ of the image equals the column rank of $\mathcal{M}(T)$, that is, the dimension of the span of its columns.

*Proof.* The coordinate map from $W$ to $\mathbb{F}^m$, $w \mapsto [w]_C$, is linear and injective. It carries $\operatorname{im} T = \operatorname{span}(Tv_1,\dots,Tv_n)$ onto the span of the vectors $[Tv_k]_C$, and these vectors are exactly the columns of $\mathcal{M}(T)$. Let $u_1,\dots,u_r$ be a basis of the image $\operatorname{im} T$, so $r = \dim \operatorname{im} T$. By Lecture 3's proposition "Action on lists", the vectors $[u_1]_C,\dots,[u_r]_C$ span the column span and, since the map is injective, are linearly independent. Hence they form a basis of the column span, and the column rank equals $r = \dim \operatorname{im} T$. ∎

**Remark 4.14.** The image $\operatorname{im} T$ does not depend on the bases, so neither does the column rank. This result also connects map rank with Gaussian elimination. Reduce a matrix $A$ to its reduced row-echelon form $R$. Every elementary row operation is invertible, so the equations $Ax = 0$ and $Rx = 0$ have the same solutions. Hence the columns of $A$ and of $R$ satisfy the same linear relations, and their column ranks agree. The pivot columns of $R$ are linearly independent, and every column without a pivot is a linear combination of them. So if $R$ has $r$ pivots, the column rank of $A$ equals $r$.

The same number $r$ is also the *row rank*, the dimension of the span of the rows. Each row operation replaces rows by linear combinations of rows and is invertible, so the span of the rows does not change. The nonzero rows of $R$ are linearly independent, since each has a pivot where the other rows have zeros. Hence the row rank of $A$ is also $r$: row rank equals column rank. So Gaussian elimination computes the map rank $\dim \operatorname{im} T$, which does not depend on the bases.

**Example 4.15 (Dimension of a map space).** For example, $\dim \mathcal{L}(\mathbb{R}^3,\mathbb{R}^2) = 6$. A basis of this space consists of the six maps whose matrices are matrix units: one of the six entries is 1 and all the others are zero. ◁

## 3 Invertibility and isomorphisms

**Definition 4.16 (Invertible map, isomorphism).** A linear map $T \in \mathcal{L}(V,W)$ is called *invertible* if there is a linear map $S \in \mathcal{L}(W,V)$ such that $ST = I_V$ and $TS = I_W$. Such an $S$ is unique; we write it $T^{-1}$. An invertible linear map is called an *isomorphism*. The spaces $V, W$ are called *isomorphic* (written $V \cong W$) if there is an isomorphism between them.

The next proposition says that a linear map is invertible if and only if it is bijective. After it we prove the main fact of this section: whether two finite-dimensional spaces are isomorphic is decided by a single number — the dimension.

**Proposition 4.17 (Invertible = bijective).** A linear map $T \in \mathcal{L}(V,W)$ is invertible if and only if it is both injective and surjective. In that case its inverse function is automatically linear.

*Proof.* An invertible map has a two-sided inverse, so it is a bijection. Conversely, let $T$ be a bijection and $T^{-1}$ its inverse function. Take any two vectors of $W$ and write them as $w_i = Tv_i$ (possible because $T$ is surjective). Then for any $a \in \mathbb{F}$ we have

$$T^{-1}(w_1 + aw_2) = T^{-1}\big(T(v_1 + av_2)\big) = v_1 + av_2 = T^{-1}w_1 + a\,T^{-1}w_2,$$

so $T^{-1}$ is linear. By construction it is a two-sided inverse. ∎

**Theorem 4.18 (Dimension classifies spaces).** Two finite-dimensional vector spaces over $\mathbb{F}$ are isomorphic if and only if their dimensions agree.

*Proof.* Suppose $T\colon V \to W$ is an isomorphism. Then $\ker T = \{0\}$ and $\operatorname{im} T = W$, so by rank–nullity $\dim V = \dim \operatorname{im} T = \dim W$.

Conversely, let $\dim V = \dim W = n$. Choose a basis $v_1, \dots, v_n$ of $V$ and a basis $w_1, \dots, w_n$ of $W$. Let $T$ be the linear map with $Tv_k = w_k$; in Lecture 3 we saw that such a map exists. It is surjective because the vectors $w_k$ span $W$. It is also injective because these vectors are linearly independent (so $\ker T = \{0\}$). Hence $T$ is an isomorphism. ∎

**Corollary 4.19 (The coordinate isomorphism).** If $\dim V = n$, the coordinate map $v \mapsto [v]_{\mathcal{B}}$ is an isomorphism $V \cong \mathbb{F}^n$. Thus every $n$-dimensional space over $\mathbb{F}$ is, in effect, a copy of $\mathbb{F}^n$. For example, $\mathcal{P}_m(\mathbb{F}) \cong \mathbb{F}^{m+1}$ and $M_{m\times n}(\mathbb{F}) \cong \mathbb{F}^{mn}$.

**Example 4.20 (An explicit isomorphism).** The space $\operatorname{Sym}_{2\times 2}(\mathbb{R})$ of symmetric $2 \times 2$ real matrices and the space $\mathbb{R}^3$ both have dimension 3, so these spaces are isomorphic. We can write an explicit isomorphism as

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix} \mapsto (a, b, c).$$

This is exactly the coordinate map for the basis

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

For the purposes of linear algebra, spaces that look quite different are the same once their dimensions match. ◁

**Example 4.21 (Transporting a question to $\mathbb{F}^n$).** Do the polynomials $1+t,\ t+t^2,\ 1+t^2$ form a basis of $\mathcal{P}_2(\mathbb{R})$? Use the coordinate isomorphism $\mathcal{P}_2(\mathbb{R}) \cong \mathbb{R}^3$ for the standard basis $1, t, t^2$. It sends these polynomials to the vectors $(1,1,0), (0,1,1), (1,0,1)$. Writing them as the columns of a matrix, we compute the determinant:

$$\det\begin{pmatrix}1 & 0 & 1\\ 1 & 1 & 0\\ 0 & 1 & 1\end{pmatrix} = 2 \neq 0.$$

Hence these vectors are linearly independent in $\mathbb{R}^3$. An isomorphism preserves linear independence and spanning, so these three polynomials also form a basis of $\mathcal{P}_2(\mathbb{R})$. A single numerical determinant has settled an abstract question. ◁

In the language of matrices: $T$ is invertible if and only if its matrix $\mathcal{M}(T)$ is an invertible square matrix. In that case $\mathcal{M}(T^{-1}) = \mathcal{M}(T)^{-1}$. This follows directly from Theorem 4.9 applied to $T^{-1}T = I$.

**Example 4.22 (Invertibility through the matrix).** The matrix of the operator $T(x,y) = (x+2y,\ 3x+4y)$ on $\mathbb{R}^2$ is $A = \begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$. Its determinant is $\det A = -2 \neq 0$, so $T$ is invertible. The inverse matrix is

$$A^{-1} = \frac{1}{-2}\begin{pmatrix}4 & -2\\ -3 & 1\end{pmatrix} = \begin{pmatrix}-2 & 1\\ \frac{3}{2} & -\frac{1}{2}\end{pmatrix}.$$

It is the matrix of the operator $T^{-1}$, that is,

$$T^{-1}(x,y) = \Big(-2x+y,\ \frac{3}{2}x - \frac{1}{2}y\Big).$$

An operator is invertible if and only if its matrix is invertible. ◁

**Example 4.23 (A non-invertible operator).** The differentiation operator $D$ on $\mathcal{P}_3(\mathbb{R})$ is not invertible. The first column of its matrix

$$\mathcal{M}(D) = \begin{pmatrix}0 & 1 & 0 & 0\\ 0 & 0 & 2 & 0\\ 0 & 0 & 0 & 3\\ 0 & 0 & 0 & 0\end{pmatrix}$$

is zero, so $\det \mathcal{M}(D) = 0$. Indeed, the kernel $\ker D$ consists of the constant polynomials $p(t) = a$. No change of basis can repair this: invertibility is a property of the operator, not of its matrix. ◁

## 4 Change of basis and similarity

The matrix of a map depends on the chosen bases. We now determine exactly how the matrix of an operator $T \in \mathcal{L}(V)$ changes when we pass from one basis of $V$ to another. Let $\mathcal{B} = (v_1, \dots, v_n)$ be the "old" basis and $\mathcal{B}' = (v_1', \dots, v_n')$ the "new" one.

**Definition 4.24 (Change-of-basis matrix).** The *change-of-basis matrix* $P$ is the $n \times n$ matrix whose $k$-th column holds the coordinates of the new basis vector $v_k'$ in the old basis $\mathcal{B}$. Equivalently, $P = \mathcal{M}(I, \mathcal{B}', \mathcal{B})$.

The matrix $P$ is the matrix of the identity map, so by Theorem 4.6, for every vector $v$,

$$[v]_{\mathcal{B}} = P\,[v]_{\mathcal{B}'}. \tag{4.2}$$

So the change-of-basis matrix $P = \mathcal{M}(I, \mathcal{B}', \mathcal{B})$ carries coordinate vectors from the new basis $\mathcal{B}'$ to the old basis $\mathcal{B}$. The identity map is invertible, and so is $P$: its inverse $P^{-1}$ carries coordinate vectors from the old basis to the new one, $[v]_{\mathcal{B}'} = P^{-1}[v]_{\mathcal{B}}$.

**Example 4.25 (Converting coordinates).** Let $\mathcal{B}$ be the standard basis of $\mathbb{R}^2$ and $\mathcal{B}' = \big((1,1),(1,-1)\big)$. Writing the new basis vectors as columns gives

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Let the coordinates of a vector $v$ in the new basis be $[v]_{\mathcal{B}'} = (3,1)$. Its old coordinates are

$$[v]_{\mathcal{B}} = P\,[v]_{\mathcal{B}'} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix},$$

that is, $v = (4,2)$. Check: $3(1,1) + 1(1,-1) = (4,2)$. Note that when we multiply by matrices, we write coordinate vectors as columns. ◁

**Theorem 4.26 (Similarity).** Let an operator $T \in \mathcal{L}(V)$ have matrices $A = \mathcal{M}(T, \mathcal{B})$ and $A' = \mathcal{M}(T, \mathcal{B}')$, and let $P = \mathcal{M}(I, \mathcal{B}', \mathcal{B})$ be the change-of-basis matrix of Definition 4.24. Then

$$A' = P^{-1}AP.$$

*Proof.* Take any vector $v$. Applying (4.2) and Theorem 4.6 repeatedly, we get

$$A'[v]_{\mathcal{B}'} = [Tv]_{\mathcal{B}'} = P^{-1}[Tv]_{\mathcal{B}} = P^{-1}A\,[v]_{\mathcal{B}} = P^{-1}AP\,[v]_{\mathcal{B}'}.$$

This equality holds for all coordinate vectors $[v]_{\mathcal{B}'}$, so $A' = P^{-1}AP$ (Figure 2). ■

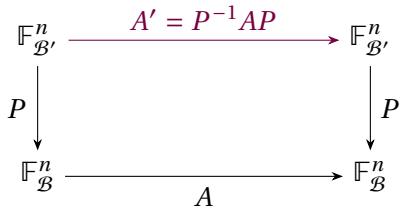



Figure 2: Similarity. To apply $T$ in the new basis, we first convert the coordinates to the old basis ($P$), then multiply by $A$, and convert back to the new basis ($P^{-1}$).

**Definition 4.27 (Similar matrices).** Square matrices $A$ and $A'$ are *similar* if there is an invertible matrix $P$ such that $A' = P^{-1}AP$. Similar matrices are exactly the matrices of one operator in different bases.

**Proposition 4.28 (Similarity invariants).** Similarity is an equivalence relation, and similar matrices have the same trace and the same determinant.

*Proof.* Reflexivity ($P = I$), symmetry ($A = PA'P^{-1}$) and transitivity (multiply the change-of-basis matrices) are checked directly. For the trace, use the identity $\operatorname{tr}(XY) = \operatorname{tr}(YX)$:

$$\operatorname{tr}(P^{-1}AP) = \operatorname{tr}\big(A(PP^{-1})\big) = \operatorname{tr} A.$$

For the determinant,

$$\det(P^{-1}AP) = \det(P)^{-1}\det(A)\det(P) = \det A.$$ ■

Hence trace and determinant are properties of the *operator*, not of any particular matrix of it. In Lecture 5 we will see that the same holds for eigenvalues: similar matrices have the same eigenvalues. This is why diagonalization makes sense.

**Remark 4.29 (Operators and general maps).** Similarity compares two matrices of one *operator*, so the same basis is used on the domain and on the codomain. For a general map $T \in \mathcal{L}(V, W)$ the bases can be changed independently: by a matrix $P$ on $V$ and by a matrix $Q$ on $W$. The corresponding formula is then $A' = Q^{-1}AP$, and such matrices are called *equivalent*. Equivalence is weaker than similarity: the matrix of every rank-$r$ map is equivalent to the block matrix with an $r \times r$ identity matrix in the top-left corner and zeros elsewhere. This normal form can be reached with facts we have already proved. Choose a basis of $V$ whose last vectors span $\ker T$. A basis of $W$ consists of the images of the first basis vectors, extended to a basis. We do not carry out this construction here. Similarity preserves more information — trace, determinant, eigenvalues. These quantities make sense only when the domain and the codomain coincide.

**Example 4.30 (The matrix of a reflection in a suitable basis).** Let $T \in \mathcal{L}(\mathbb{R}^2)$ be the reflection across the line $y = x$, that is, $T(x, y) = (y, x)$. Its matrix in the standard basis is $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Choose the new basis $\mathcal{B}' = \big((1,1), (1,-1)\big)$. For it,

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad \text{and} \quad P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then

$$A' = P^{-1}AP = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The vectors of the new basis $\mathcal{B}'$ suit the reflection: the first lies on the mirror line, and the second is perpendicular to it. So in this basis the matrix is diagonal, with the obvious factors $+1$ and $-1$ on its diagonal. This is the first example of diagonalization: a well-chosen basis shows what the operator actually does. ◁

**Example 4.31 (The same matrix, straight from the definition).** We can also obtain the matrix $\mathcal{M}(T, \mathcal{B}')$ without the similarity formula. We express the images of the new basis vectors in terms of the same basis:

$$\begin{aligned} T(1,1) = (1,1) &= 1 \cdot (1,1) + 0 \cdot (1,-1), \\ T(1,-1) = (-1,1) &= 0 \cdot (1,1) + (-1) \cdot (1,-1). \end{aligned}$$

The coefficient vectors $(1,0)$ and $(0,-1)$ are the columns of the matrix, so $\mathcal{M}(T, \mathcal{B}') = \operatorname{diag}(1,-1)$. Both routes give the same result, as Theorem 4.26 says they must. ◁

**Example 4.32 (Similarity links the two matrices of $D$).** Return to the differentiation operator $D$ on $\mathcal{P}_2(\mathbb{R})$. Take the standard basis $\mathcal{B} = (1, t, t^2)$ and the basis $\mathcal{B}' = (1,\ 1+t,\ 1+t+t^2)$ from Example 4.5. The columns of the change-of-basis matrix hold the coordinates of the vectors of $\mathcal{B}'$ in the basis $\mathcal{B}$:

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad P^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

In the standard basis,

$$A = \mathcal{M}(D, \mathcal{B}) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Multiplying the matrices gives

$$P^{-1}AP = \begin{pmatrix} 0 & 1 & -1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix} = \mathcal{M}(D, \mathcal{B}').$$

The result agrees with the matrix we computed directly in Example 4.5, as Theorem 4.26 states. ◁

**Example 4.33 (Invariants in action).** Return to the reflection of Example 4.30. The trace of the matrix $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ is $\operatorname{tr} A = 0$, and its determinant is $\det A = -1$. The trace of the diagonal matrix $A' = \operatorname{diag}(1, -1)$ is $\operatorname{tr} A' = 1 + (-1) = 0$, and its determinant is $\det A' = -1$. The invariants agree, as Proposition 4.28 promises. In Lecture 5 we will see that the eigenvalues of this reflection are $+1$ and $-1$ — exactly the diagonal entries that the well-chosen basis revealed. ◁

## The map, not the matrix

A matrix is only a record of a linear map in a chosen basis, not the map itself. Quantities that do not change under a change of basis — rank, trace, determinant, and soon eigenvalues — are properties of the operator itself. Everything else depends on the chosen coordinates. In the coming lectures we will use this freedom: we will look for a basis in which the matrix is diagonal, so that the action of the operator becomes clear. In quantum mechanics one often passes to a basis made of states of definite energy. In this basis the evolution operator $e^{-i\hat{H}t/\hbar}$ is diagonal, so each basis state only acquires a phase factor as time goes on.

**Remark 4.34 (Forward preview: density matrices).** In later lectures we study finite-dimensional quantum models. In them a state is described by a density matrix $\rho$ — a positive operator of trace one — and an observable by a self-adjoint operator $A$. Under a change of basis, the matrices of both operators change in the same way:

$$\rho' = P^{-1}\rho P \quad \text{and} \quad A' = P^{-1}AP.$$

Therefore

$$\operatorname{tr}(\rho' A') = \operatorname{tr}(P^{-1}\rho AP) = \operatorname{tr}(\rho A).$$

Thus the expectation value $\langle A \rangle = \operatorname{tr}(\rho A)$ does not depend on the chosen coordinates. It is a property of the physical quantity, not of the way we write it down. Lecture 10 gives the physical definitions and hypotheses; here we have only applied the change-of-basis formula.

## Summary of main concepts

Once bases of $V$ and $W$ are fixed, we can write each map $T \in \mathcal{L}(V, W)$ as a matrix $\mathcal{M}(T)$. Its columns are the images of the basis vectors, written in coordinates, and $[Tv]_C = \mathcal{M}(T)[v]_{\mathcal{B}}$. Composition of maps corresponds to matrix multiplication; that is exactly why matrices are multiplied by the "row times column" rule. The space of matrices is isomorphic to the space of maps: $\mathcal{L}(V, W) \cong \mathbb{F}^{m\times n}$ and $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$. A map is an isomorphism if and only if it is bijective. Finite-dimensional spaces are isomorphic if and only if their dimensions agree, so every $n$-dimensional space is, in effect, a copy of $\mathbb{F}^n$. Under a change of basis, an operator's matrix $A$ becomes the similar matrix $P^{-1}AP$. Trace and determinant are similarity invariants, so they are properties of the operator itself.

- **Matrix of a map**—its columns are the images of the basis vectors, written in coordinates; the matrix depends on the basis.

- **Composition → multiplication**—this is where the row-times-column rule comes from; $\mathcal{L}(V,W) \cong \mathbb{F}^{m\times n}$.
- **Isomorphism**—a bijective linear map; finite-dimensional spaces are isomorphic $\iff$ their dimensions agree.
- **Change of basis / similarity**—$A \mapsto P^{-1}AP$; trace and determinant are invariant, so they belong to the operator.